# Jinwon Lee

Agent Platforms · Applied AI · Reliability

Website

https://jinwonlee.github.io/

This portfolio summarizes the agent platform and reliability engineering work that takes enterprise and user-facing applications from requirements through implementation, rollout, and production stabilization. Each project is organized around the problem context, my role, engineering and delivery focus, outcomes, and capability signals.

### Agent Platforms

Agent runtime integration, self-service application delivery, and existing LLM Gateway API connectivity

### Governed Delivery

Authentication, authorization, security controls, automatic domains, and enterprise governance

### Reliability

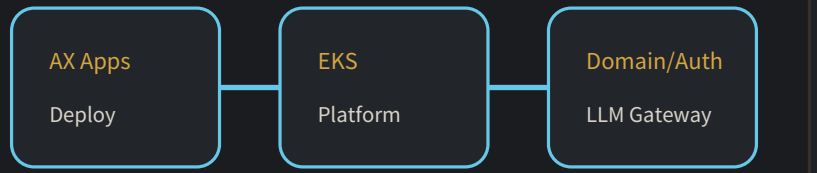
Observability, incident response, rollout readiness, and production stabilization

### Cloud Runtime

AWS EKS, Azure, OpenShift, Kubernetes, Helm, ArgoCD, and GitOps

# Project Details

Problem, role, engineering/delivery focus, outcome, and capability point

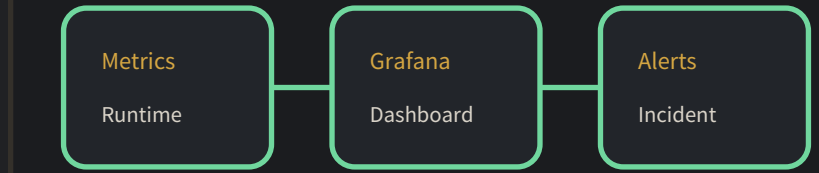



## 01. Internal AX Agent Platform

AWS EKS-based platform for deploying and operating AX applications with automated domains, access control, governance, and LLM Gateway integration.

| | |
|---|---|
| Problem | Enable teams to deploy AX applications quickly while enforcing enterprise access control, security, and AI governance. |
| Role | Owned end-to-end platform delivery from architecture and implementation through deployment and operation, including authentication, authorization, security, and internal governance. |
| Engineering | AWS EKS, AX Agent runtime adaptation, automated domain provisioning, access control, policy enforcement, and client integration with the existing LLM Gateway. |
| Outcome | Created a self-service path for governed AX application deployment with automatic domains and authorized-only access. |
| Capability | Experience building a governed self-service agent platform with runtime compatibility, access control, and enterprise LLM integration. |

AWS · EKS · Kubernetes · AuthN/AuthZ · Governance




## 02. Centralized Monitoring & Observability Stack

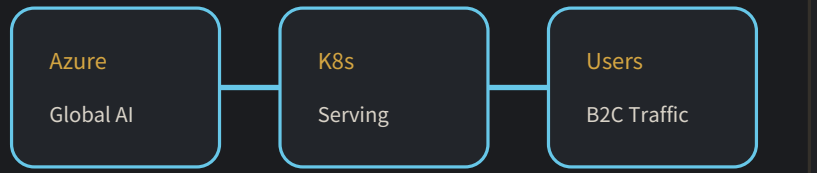
Centralized observability stack for system metrics, container runtime metrics, dashboards, and alerts.

| | |
|---|---|
| Problem | Production services needed shared visibility into system behavior and container runtime health. |
| Role | Built the monitoring path and dashboards used for platform and service operations. |
| Engineering | Prometheus metrics, Grafana dashboards, OpenTelemetry signals, Loki logging, and alert rules. |
| Outcome | Improved operational visibility for incident detection and runtime analysis. |
| Capability | Experience using metrics, logs, and traces for incident detection and runtime analysis. |

Grafana · Prometheus · OpenTelemetry · Loki

# Project Details

Problem, role, engineering/delivery focus, outcome, and capability point




## 03. Global B2C Generative AI Service

Microsoft Azure-based B2C generative AI service expanded to the UK with initial production operations and incident response.

| | |
|---|---|
| Problem | Expand a consumer-facing generative AI service to the UK and stabilize its initial production operations. |
| Role | Delivered platform/SRE responsibilities for UK rollout readiness, initial production operations, incident response, and early stabilization. |
| Engineering | Azure and Kubernetes operations, rollout readiness, production stabilization, and incident response. |
| Outcome | Delivered UK rollout readiness and early production stabilization, completing the engagement in June 2026. |
| Capability | Experience delivering UK rollout readiness, production stabilization, and incident response for a consumer generative AI service. |

Azure · Kubernetes · Generative AI · SRE

## 04. Company-wide AI Chatbot Platform (30K Users)

Enterprise AI chatbot platform serving 30,000 employees with Kubernetes-based application delivery, reliability, and data protection.

| | |
|---|---|
| Problem | Provide a reliable internal AI assistant platform for tens of thousands of employees. |
| Role | Led platform and application reliability across development, staging, and production, connecting releases with security and operational requirements. |
| Engineering | Kubernetes operations, Helm/Helmfile IaC, AI service delivery, authentication, and data protection compliance. |
| Outcome | Supported a company-wide AI service with production-grade operating standards. |
| Capability | Experience leading application delivery and reliability for tens of thousands of internal users. |

Kubernetes · Internal Cluster · Application Delivery

# Project Details

Problem, role, engineering/delivery focus, outcome, and capability point

## 05. GitOps CI/CD Pipeline with ArgoCD

GitOps-based delivery pipelines for containerized AI/platform services with Python automation and DevSecOps checks.

| | |
|---|---|
| Problem | Containerized AI/platform services needed repeatable, auditable, and secure deployment workflows. |
| Role | Designed delivery automation and connected AI service deployment pipelines with security verification steps. |
| Engineering | ArgoCD GitOps, GitHub Actions, Docker image handling, Kubernetes deployment, and Python glue automation. |
| Outcome | Improved deployment consistency and reduced reliance on manual release operations. |
| Capability | Experience connecting deployment consistency, security checks, and operations automation for containerized services. |

ArgoCD · GitHub Actions · Docker · Kubernetes

## 06. Kubernetes Cluster Automation

One-click Kubernetes cluster provisioning for repeatable developer and test environments.

| | |
|---|---|
| Problem | Cloud developers needed repeatable Kubernetes environments without manual setup overhead. |
| Role | Designed and implemented provisioning automation for Kubernetes clusters and shared platform modules. |
| Engineering | Bash-based provisioning and automation, reusable cluster modules, and developer environment standardization. |
| Outcome | Reduced manual operations and improved environment consistency for cloud development workflows. |
| Capability | Experience standardizing infrastructure and automating repeatable development and test environments. |

Kubernetes · Bash

# Project Details

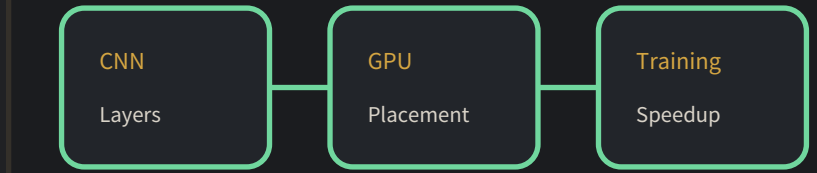
Problem, role, engineering/delivery focus, outcome, and capability point

## 07. High-throughput Packet Mirroring Application

Real-time packet mirroring application that forwarded up to 130,000 packets/sec under constrained resources.

| | |
|---|---|
| Problem | Network/application teams needed real-time packet forwarding under limited system resources. |
| Role | Implemented the packet mirroring application and tuned it for resource-constrained operation. |
| Engineering | High-throughput I/O, Linux runtime behavior, resource limits, and real-time data forwarding. |
| Outcome | Delivered a system component capable of forwarding up to 130,000 packets/sec. |
| Capability | Experience optimizing Linux runtime behavior and high-throughput data movement under resource constraints. |

Linux · Networking · System Software · Performance




## 08. Distributed CNN Training System

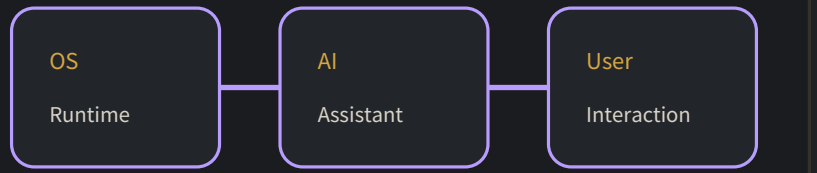
Research on accelerated CNN distributed deep learning with automatic resource-aware layer placement.

| | |
|---|---|
| Problem | Distributed CNN training suffered from network bottlenecks and inefficient layer placement. |
| Role | Contributed to research on resource-aware placement for distributed TensorFlow/CNN training. |
| Engineering | Distributed systems, model parallelism, resource-aware scheduling, TensorFlow, and GPU-aware workloads. |
| Outcome | Published research artifacts including USENIX NSDI 2019 poster and arXiv paper; reported up to 2.3x training speedup. |
| Capability | Research experience on network bottlenecks, resource placement, and performance improvement for distributed training workloads. |

TensorFlow · Distributed Systems · AI Workloads · Python

# Project Details

Problem, role, engineering/delivery focus, outcome, and capability point




## 09. Siri-like AI Assistant on TmaxOS

AI assistant with system-level integration and natural user experience.

| | |
|---|---|
| Problem | The OS environment needed an AI assistant that could connect user interaction with system-level capabilities. |
| Role | Built AI inference and preprocessing flows and integrated them with TmaxOS-level behavior. |
| Engineering | Async architecture, system integration, AI model inference, and data preprocessing. |
| Outcome | Delivered an AI assistant experience with smoother interaction through asynchronous design. |
| Capability | Experience integrating AI functionality with OS-level behavior, asynchronous processing, and user interaction. |

Python · AI/ML · TmaxOS · Async Programming

## 10. LLM-Assisted Cluster Monitoring

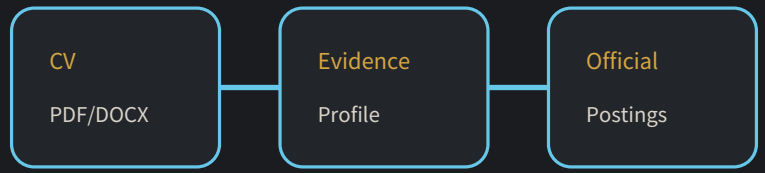
Personal project for internal cluster operations: error detection, LLM analysis, and email reports with suggested remediation steps.

| | |
|---|---|
| Problem | Cluster errors need interpretation and practical troubleshooting guidance, beyond a basic alert. |
| Role | Built the monitoring and reporting workflow as a personal project for internal cluster operations. |
| Engineering | Connected error detection, LLM-based issue analysis, and email delivery of problem summaries and remediation suggestions. |
| Outcome | Delivered email reports containing the detected problem, LLM analysis, and suggested steps for the operator to review. |
| Capability | Applies LLM analysis to cluster operations and operator-facing incident reporting. |

LLM · Cluster Monitoring · Email Alerts

# Project Details

Problem, role, engineering/delivery focus, outcome, and capability point




## 11. OfferNow

Personal project: a CV-based job recommendation service for Korean IT roles that shows the CV evidence behind every recommended official posting.

| | |
|---|---|
| Problem | Job seekers have to read many postings to see which ones actually fit their experience, and keyword matching gives no reason they can check. |
| Role | Personal project: product definition, backend, mobile and web client, data collection, and deployment. |
| Engineering | Expo/React Native client for web, iOS, and Android; FastAPI backend; PostgreSQL with row-level security and anonymous 24-hour sessions; a daily collector with a durable task queue; rule-based matching; provider-neutral LLM calls with usage limits and cost logging; GitHub Actions CI. |
| Outcome | Deployed at heyoffernow.ai behind an operator-only gate ahead of a beta; collects official postings from 50+ companies. |
| Capability | Builds a full AI product end to end, from data collection and privacy controls to LLM integration, cost limits, and deployment. |

Expo FastAPI PostgreSQL LLM Docker

# Links and References

Detailed projects and the latest CV are available on the online portfolio.


Website

https://jinwonlee.github.io/

GitHub

https://github.com/JINwonLEE

LinkedIn

https://www.linkedin.com/in/jwl1993

Email

dlwlsdnjsehs1993@gmail.com


## Awards & Activities

- Finalist, SK hynix AI Hackathon 2026 (Nov. 2026): Advanced through two online qualifying rounds to the offline finals (individual entry).

## Publications

- Alleviating the Network Bottleneck for CNN Distributed Training through Automatic Resource-Aware Layer Placement — USENIX NSDI 2019 (2019)
- Accelerated Training for CNN Distributed Deep Learning through Automatic Resource-Aware Layer Placement — arXiv 2019 (2019)
- Improving Performance of Distributed TensorFlow using CNN Characteristics Exploiting Model Parallelism — ACM EuroSys 2018 (2018)

Additional experience: contributed to LLM evaluation work for machine translation.

This document is a portfolio summary based on publicly shareable career and project information. It does not include confidential implementation details or non-public information.